

SEROM: SPEECH ENHANCEMENT WITH REPRESENTATION-OPTIMIZED SSL MODELING

Seyun Um¹, Doyeon Kim², Hong-Goo Kang², Nilesh Madhu¹

¹Department of Electrical Engineering, Helmut Schmidt University, Hamburg, Germany

²Department of Electrical and Electronic Engineering, Yonsei University, Seoul, South Korea

ABSTRACT

Recent neural speech enhancement (SE) systems leveraging pre-trained speech self-supervised learning (SSL) models have demonstrated strong performance. However, little attention has been paid to systematically analyzing how representations from different SSL layers can be utilized and how each contributes to SE performance. In this work, using early-layer SSL representations as primary acoustic features, we investigate the effects of incorporating contextual-related information extracted from deeper layer of the same SSL backbone. Experimental results reveal that compact contextual representations play a critical role in improving SE performance. Rather than relying on an embedding-level matching loss, we explicitly train the higher-layer representation with a text prediction loss, which reduces irrelevant information and encourages more informative contextual features for SE. Motivated by these findings, we propose a novel SE framework, termed *Serom*, which explicitly exploits compressed contextual representations derived from late SSL layers. The resulting representations are integrated through a decoder and converted into enhanced speech using a neural vocoder. Experimental evaluations demonstrate that the proposed approach achieves state-of-the-art performance in terms of speech quality, intelligibility, and speaker similarity.¹

Index Terms— speech enhancement, self-supervised learning

1. INTRODUCTION

Speech enhancement (SE) aims to recover clean and intelligible speech signals from noisy observations. As a front-end for real-world applications, including automatic speech recognition (ASR), hearing assistive devices, and telecommunication systems, robust SE is crucial for reliable human-machine interaction under adverse acoustic conditions [1].

Early SE relied on statistical signal processing techniques, such as Wiener filtering [2] and minimum mean squared error (MMSE)-based methods [3], which assume noise stationarity and simplified signal models. Although computationally efficient, they often exhibit limited performance in highly non-stationary and complex acoustic environments, leaving

residual artifacts and speech distortion, motivating the shift toward data-driven approaches with the emergence of deep neural networks (DNNs).

Neural SE has evolved along several directions: time–frequency mapping or mask estimation with convolutional and recurrent networks [4, 5], time-domain end-to-end models that jointly handle magnitude and phase [6], and more recent complex-valued representations [7], attention mechanisms [8], and generative frameworks such as GANs and diffusion models [9, 10]. Despite these advances, most methods rely on supervised learning over paired noisy–clean data, limiting generalization to unseen conditions.

To address these challenges, recent studies increasingly adopt self-supervised learning (SSL), learning speech representations from large-scale unlabeled corpora. Pre-trained SSL models encode rich hierarchical information [11, 12]—acoustic, phonetic, and speaker-related—making them well suited to SE, and several SSL-based methods [13, 14] show improved robustness and generalization in low-resource and mismatched conditions. Recent work also explores token-based generative frameworks using discrete semantic, acoustic, or codec tokens [15, 16].

However, most SSL-based SE methods exploit only limited aspects of SSL representations, typically the final hidden layer or aggregations of multiple layers [17, 18], underusing their hierarchical nature, where layers range from low-level acoustic to high-level semantic and speaker information. As a result, it remains unclear how to design acoustic representations that preserve fine-grained detail while incorporating contextual information beneficial for SE.

In this paper, we propose *Serom*, a novel SE framework that more effectively exploits hierarchical SSL representations. Specifically, we decompose the multi-layer SSL features into foundational acoustic representations extracted from early layers and complementary contextual representations derived from deeper layers. These complementary representations are incorporated as structured conditioning information to guide the enhancement process, enabling more accurate and robust speech reconstruction under challenging acoustic conditions. All representations required for enhancement are obtained from a single SSL backbone, with different layers assigned distinct functional roles according to their

¹Our demo page: <https://sam-0927.github.io/serom/>

representational characteristics.

We observed that contextual information extracted from deeper layers plays a crucial role in improving enhancement performance. Furthermore, projecting contextual representations into a low-dimensional feature space improves performance, likely due to the implicit denoising effect of compact representations.

The main contributions of this paper are summarized as follows.

- We propose Serom, a novel speech enhancement framework that combines foundational acoustic representations from an early SSL layer with compressed contextual representations derived from a deeper layer of the same SSL backbone.
- Experimental results show that the proposed approach achieves state-of-the-art performance in speech intelligibility, perceptual quality, and speaker similarity.

2. PROPOSED METHOD

Figure 1 illustrates the overall architecture of Serom. Given noisy speech, Serom extracts two representations from a single pre-trained WavLM backbone, an acoustic feature and a context embedding. The acoustic feature serves as the primary representation to be enhanced, while the context embedding is utilized as an auxiliary condition. These two features are fused by a decoder to generate an intermediate mel-spectrogram representation, which is subsequently converted into enhanced speech by a vocoder. The pre-trained WavLM and decoder are first jointly trained, after which the vocoder is trained while the remaining modules are kept frozen.

2.1. Feature Extractor

The acoustic feature, which serves as the primary representation to be enhanced, is extracted from the first transformer layer of WavLM. As discussed in Section 1, we employ the early-layer representation as the main feature because it preserves the explicit acoustic structure more clearly than the final-layer output. Since this representation is obtained after passing through the convolutional frontend, it retains sufficiently rich acoustic information for neural signal generation. The first-layer output is then transformed into latent representations through a projection layer.

To improve the intelligibility of the output signal by incorporating context-related information, we utilize the representation from the 24th layer of WavLM, denoted as H_{24} . The motivation behind this design is that deeper SSL representations tend to encode richer linguistic information, which can implicitly provide text-related conditioning for the enhancement process. To more explicitly inject textual information into this representation, we apply Connectionist Temporal Classification (CTC) [19] supervision. Specifically, H_{24} is first projected by f_{ctc} to match the vocabulary size, and the

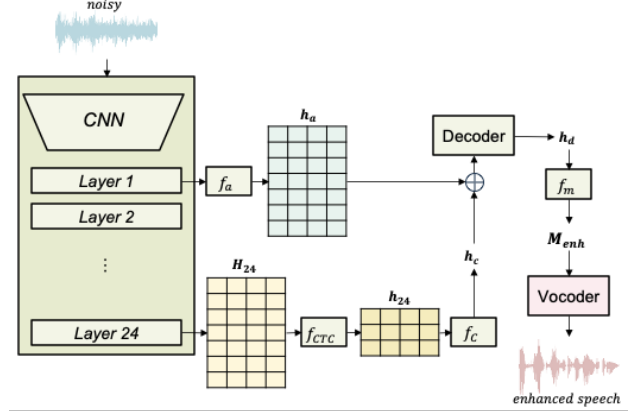


Fig. 1. Overall of Serom. The green modules are trained first, followed by the pink module.

resulting logits, h_{24} , are used to predict the target text using the CTC loss. Subsequently, H_{24} is projected again through f_c to match the dimensionality of the acoustic feature, yielding the final context embedding used for conditioning.

$$L_{CTC} = CTC(h_{24}), \quad (1)$$

$$h_c = f_c(h_{24}), \quad h_{24} = f_{ctc}(H_{24}). \quad (2)$$

2.2. Speech Generator

The decoder integrates the acoustic feature and context embedding to generate an intermediate feature for waveform synthesis. Specifically, the two features are first summed and combined with positional embeddings. The resulting representation is then fed into a transformer encoder consisting of four blocks, each equipped with eight self-attention heads and a feed-forward dimension of 2048:

$$h_d = \text{Dec}(h_a + h_c + E^{\text{pos}}), \quad (3)$$

where $h_d \in R^{T \times 1024}$. To regularize the decoder output toward clean speech, we employ a prediction head, f_m , to map h_d to an 80-dimensional mel-spectrogram, denoted as M_{enh} , and optimize it using an ℓ_1 loss.

$$L_{\text{mel}} = \|M_{\text{enh}} - M_{\text{clean}}\|_1, \quad (4)$$

where M_{clean} represents the target clean mel-spectrogram.

The final enhanced waveform is generated from the decoder output using a neural vocoder. In our implementation, we adopt Vocos [20], a Fourier-based neural vocoder that synthesizes waveform samples from the decoder feature. The vocoder is trained while keeping the remaining modules frozen, following the original loss configuration.

3. EXPERIMENTS

3.1. Experimental setup

Dataset. We used the 16-kHz LibriSpeech corpus for training and evaluation. Clean speech from `train-clean-360`

was mixed with noise samples from the DNS Challenge 2020 dataset [21]. The SNR was sampled from $[-5, 15]$ dB, with the sampling distribution adjusted such that 0 and -5 dB accounted for 50% of the training data. Room reverberation was applied with a probability of 50% using SLR26 RIRs [22]. For evaluation, noisy test utterances were generated from `test-clean` using noise samples from the DNS Challenge test set and SLR28 RIRs [22]. The evaluation data were divided into fixed SNR levels of $[-5, 0, 5, 10, 15]$ dB. We additionally evaluated the proposed model on the DNS Challenge test set to further assess its generalization ability under unseen test conditions.

We compare the proposed model with three representative baseline systems: LCTGAN [10], CVAE [7]², and PASE [14]. LCTGAN, CVAE, and PASE contain 135K, 9.77M, and 382M parameters, respectively, while the proposed model contains 433.7M parameters. To ensure fair comparison, all baseline models are trained on the same dataset using their official implementation and training protocol.

Implementation Details. We used a frozen pre-trained WavLM-Large backbone³ and set the CTC vocabulary size to 29. The 1024-dimensional decoder output was cropped to 4 seconds and fed into the vocoder. For L_{mel} , we used 80-dimensional mel-spectrograms with a hop size 320, FFT size of 1280, and window size of 1280. The weighting factors for L_{CTC} and L_{mel} were set to 2 and 5, respectively. The vocoder configuration followed that of Vocos. We employed a multi-resolution discriminator [23] and a multi-band multi-scale STFT discriminator, and optimized the vocoder using feature matching, GAN, and reconstruction losses with weights of 2, 1, and 15, respectively [24].

We trained the model using AdamW with $\beta_1 = 0.8$, $\beta_2 = 0.99$, and weight decay 0.01. The learning rate was linearly warmed up to 2.0×10^{-4} during the first epoch and then decayed using cosine annealing until epoch 25.

Evaluation Metrics. We compared the proposed model with existing speech enhancement systems. For intelligibility evaluation, we measured character error rate (CER) and word error rate (WER) using OWSM v3.1 [25]. To assess speaker similarity, we extracted speaker embeddings using RawNet3 [26] and computed the cosine similarity between the enhanced speech and the corresponding clean reference.

3.2. Experimental results

As shown in Table 1, the proposed model achieves the highest speaker encoder cosine similarity (SECS) scores under reverberant conditions when the SNR is lower than or equal to 0 dB. These results demonstrate that the proposed model can enhance severely degraded speech while preserving speaker-related information more effectively than the baseline systems. Although LCTGAN and CVAE achieve relatively high SECS scores under certain conditions, they exhibit poor in-

Table 1. SECS results evaluated on the LibriSpeech.

Model	w/ Reverb.					w/o Reverb.				
	SNR					SNR				
	-5	0	5	10	15	-5	0	5	10	15
PASE	0.57	0.63	0.68	0.71	0.74	0.66	0.72	0.77	0.80	0.82
LCT-GAN	0.57	0.65	0.71	0.75	0.78	0.73	0.82	0.88	0.92	0.95
CVAE	0.58	0.65	0.72	0.77	0.80	0.75	0.83	0.89	0.93	0.96
Serom (Ours)	0.67	0.73	0.77	0.79	0.81	0.76	0.81	0.84	0.86	0.88
Serom w/o L_{CTC}	0.68	0.74	0.78	0.81	0.83	0.77	0.81	0.85	0.87	0.89
Serom w/o Decoder	0.62	0.68	0.73	0.76	0.78	0.71	0.76	0.80	0.83	0.84
Serom w/o h_c	0.16	0.19	0.23	0.24	0.26	0.31	0.37	0.43	0.48	0.51
Serom w/o h_a	0.23	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.24	0.23

Table 2. UTMOS results measured on the LibriSpeech.

Model	w/ Reverb.					w/o Reverb.				
	SNR					SNR				
	-5	0	5	10	15	-5	0	5	10	15
PASE	3.03	3.30	3.43	3.48	3.49	3.32	3.50	3.58	3.60	3.60
LCT-GAN	1.50	1.65	1.93	2.07	2.21	2.36	2.84	3.33	3.66	3.88
CVAE	1.40	1.50	1.70	1.86	2.00	2.38	2.89	3.40	3.73	3.94
Serom (Ours)	3.56	3.79	3.86	3.86	3.86	3.76	3.85	3.84	3.83	3.81
Serom w/o L_{CTC}	3.58	3.81	3.88	3.89	3.88	3.78	3.85	3.85	3.84	3.81
Serom w/o Decoder	3.35	3.66	3.79	3.83	3.83	3.69	3.86	3.90	3.91	3.91
Serom w/o h_c	1.31	1.29	1.27	1.26	1.25	1.25	1.25	1.26	1.28	1.31
Serom w/o h_a	1.81	1.89	1.88	1.86	1.84	1.76	1.75	1.70	1.65	1.62

telligibility and speech quality, suggesting that their high speaker similarity mainly results from insufficient enhancement rather than successful speech reconstruction.

Moreover, the proposed model achieves the best UTMOS scores in most conditions. The improvement is particularly pronounced under harsh acoustic environments, where the baseline systems experience substantial quality degradation. We attribute these gains to the proposed context supervision, which directly constrains the context embedding used for mel-spectrogram prediction and encourages each SSL layer to assume a more clearly defined role. As a result, the model can provide more reliable contextual guidance to the decoder and generate enhanced speech with higher perceptual quality.

A similar trend is observed on the DNS Challenge test set. The proposed model achieves higher perceptual quality while maintaining competitive speaker similarity under unseen test conditions. Although LCTGAN and CVAE obtain higher SECS scores, their relatively low speech quality indicates that the high speaker similarity does not necessarily stem from successful enhancement. In contrast, the proposed model improves perceptual quality while preserving speaker-related information, demonstrating stronger generalization capability under unseen acoustic conditions. For intelligibility evaluation, the proposed model achieves the lowest CER and WER among all comparison systems. Compared with PASE, which also separates SSL representations into multiple streams, the proposed model provides consistent improvements. This suggests that explicitly supervising the context representation with a text prediction objective offers more effective linguistic guidance for speech enhancement.

3.3. Effect of dimension reduction

We visualize the distributions of the final-layer representation H_{24} , and its dimension-reduced counterpart, h_{24} , using

²We train a teacher model.

³<https://huggingface.co/microsoft/wavlm-large>

Table 3. DNSMOS results evaluated on the LibriSpeech. Each entry denotes OVRL/SIG/BAK.

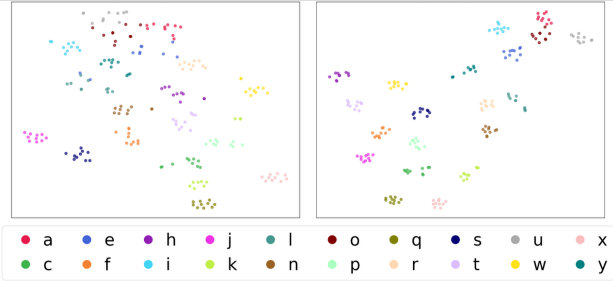
Model	w/ Reverb.					w/o Reverb.				
	SNR					SNR				
	-5	0	5	10	15	-5	0	5	10	15
PASE	3.21/3.55/3.91	3.26/3.59/3.96	3.29/3.60/3.99	3.30/3.61/4.01	3.30/3.61/4.01	3.23/3.56/3.94	3.26/3.58/3.98	3.28/3.58/3.99	3.28/3.59/4.00	3.28/3.58/4.00
LCT-GAN	2.14/2.42/3.70	2.27/2.56/3.74	2.40/2.69/3.80	2.46/2.76/3.83	2.50/2.80/3.84	2.69/3.00/3.82	2.90/3.21/3.90	3.08/3.37/3.98	3.18/3.46/4.02	3.25/3.52/4.05
CVAE	2.16/2.55/3.49	2.28/2.66/3.57	2.40/2.79/3.65	2.46/2.85/3.69	2.51/2.89/3.72	2.69/3.01/3.79	2.91/3.21/3.90	3.10/3.38/3.99	3.21/3.49/4.04	3.28/3.55/4.07
Serom (Ours)	3.25/3.58/3.95	3.29/3.60/3.99	3.31/3.61/4.01	3.31/3.61/4.02	3.31/3.61/4.02	3.29/3.60/3.99	3.31/3.61/4.02	3.31/3.61/4.02	3.31/3.61/4.02	3.31/3.60/4.02
Serom w/o L_{CTC}	3.24/3.58/3.93	3.30/3.61/4.00	3.31/3.61/4.01	3.31/3.61/4.02	3.31/3.61/4.02	3.30/3.61/4.00	3.31/3.61/4.01	3.31/3.61/4.02	3.31/3.61/4.02	3.31/3.60/4.02
Serom w/o Decoder	3.20/3.55/3.90	3.27/3.59/3.97	3.30/3.60/4.00	3.30/3.61/4.01	3.31/3.61/4.01	3.27/3.59/3.98	3.30/3.60/4.01	3.31/3.61/4.02	3.32/3.61/4.03	3.32/3.61/4.03
Serom w/o h_c	1.22/1.44/1.35	1.31/1.59/1.44	1.45/1.84/1.64	1.56/1.99/1.78	1.64/2.09/1.90	1.67/2.20/1.88	1.93/2.60/2.20	2.22/2.96/2.59	2.41/3.13/2.85	2.50/3.18/3.01
Serom w/o h_a	2.60/3.17/3.21	2.67/3.25/3.29	2.67/3.25/3.27	2.66/3.25/3.25	2.65/3.24/3.24	2.62/3.20/3.23	2.63/3.22/3.23	2.59/3.20/3.18	2.54/3.17/3.12	2.51/3.15/3.07

Table 4. CER and WER results evaluated on the LibriSpeech. Each entry denotes CER/WER.

Model	w/ Reverb.					w/o Reverb.				
	SNR					SNR				
	-5	0	5	10	15	-5	0	5	10	15
PASE	11.45/18.49	4.95/9.14	2.31/5.24	1.54/3.95	1.36/3.64	5.21/9.38	2.09/4.76	1.24/3.34	1.00/2.92	0.94/2.80
LCT-GAN	26.50/35.40	14.05/21.37	6.77/12.11	4.04/8.00	2.97/6.44	14.22/20.27	5.89/10.11	2.46/5.31	1.41/3.55	1.06/2.96
CVAE	20.15/29.04	10.91/17.80	5.22/9.92	3.26/6.77	2.32/5.35	11.36/17.41	4.92/8.97	2.13/4.79	1.28/3.29	1.00/2.86
Serom (Ours)	10.61/17.34	4.60/8.54	2.04/4.72	1.41/3.59	1.26/3.44	4.95/8.97	1.94/4.46	1.16/3.16	0.95/2.72	0.92/2.69
Serom w/o L_{CTC}	11.43/18.90	5.14/9.57	2.41/5.26	1.65/4.13	1.34/3.58	5.48/9.94	2.17/4.94	1.28/3.40	1.05/2.95	0.92/3.40
Serom w/o Decoder	11.31/18.47	4.98/9.29	2.33/5.27	1.58/3.99	1.36/3.62	5.30/9.64	2.10/4.80	1.30/3.45	1.03/2.94	0.95/2.77
Serom w/o h_c	94.09/90.91	86.26/90.49	79.17/85.22	74.73/81.64	70.04/77.82	64.94/73.57	50.84/61.58	35.19/46.92	24.94/35.97	18.41/28.71
Serom w/o h_a	10.79/17.78	4.86/9.29	2.34/5.54	1.62/4.36	1.44/4.00	5.19/9.73	2.21/5.33	1.44/4.03	1.17/3.51	1.14/3.50

Table 5. Objective evaluation results on the DNS Challenge test set. For DNSMOS, each entry denotes OVRL/SIG/BAK.

Testset [21]	Model	UTMOS	DNSMOS	SECS
w/ Reverb.	PASE	3.12	3.31/3.60/4.05	0.60
	LCTGAN	1.36	2.20/2.47/3.79	0.66
	CVAE	1.30	2.17/2.57/3.56	0.71
	Serom (Ours)	3.56	3.32/3.62/4.02	0.70
w/o Reverb.	PASE	3.49	3.33/3.60/4.08	0.80
	LCTGAN	3.57	3.22/3.46/4.09	0.92
	CVAE	3.59	3.20/3.45/4.06	0.92
	Serom (Ours)	3.70	3.36/3.62/4.09	0.88

**Fig. 2.** t-SNE visualization of h_{24} (left) and H_{24} (right) extracted from the LibriSpeech test set at 5 dB SNR without reverberation.

t-SNE [27]. The representation h_{24} is obtained by projecting H_{24} into the vocabulary space and training it to predict the target text labels. As shown in Fig. 2, the original representation H_{24} forms relatively dispersed clusters for each text label, whereas the projected representation h_{24} exhibits more compact and clearly separated clusters. These results suggest that the dimensionality reduction process does not eliminate essential text-related information. Instead, it suppresses redundant factors in the noisy SSL representation and encourages the context embedding to retain more explicit linguistic information.

3.4. Ablation study

Effect of acoustic and content features. We examine the roles of h_c and h_a by removing each feature during inference. As seen in Tab. 4, using only h_a notably degrades intelligibility, indicating that h_c is key for preserving linguistic content. In contrast, removing h_a lowers SECS (Tab. 1), implying h_a mainly encodes speaker-related acoustic information.

Effect of L_{CTC} and model size. We train Serom without L_{CTC} to assess its effect. This variant shows higher CER and WER, confirming that L_{CTC} improves intelligibility. We also test a decoder-free variant, which reduces the model size by about 50M parameters while retaining intermediate mel supervision. Although the full model performs slightly better, the decoder-free model shows no severe degradation and outperforms the larger model without L_{CTC} in intelligibility. This suggests that Serom’s performance mainly stems from L_{CTC} and intermediate mel-spectrogram supervision, rather than model size alone.

4. CONCLUSION

In this paper, we proposed Serom, a speech enhancement framework that leverages hierarchical representations from a pre-trained SSL model. Serom utilizes an early-layer WavLM representation as the acoustic feature and a final-layer representation as the context embedding. By applying a text prediction objective to the context representation, the proposed method preserves explicit linguistic information while suppressing redundant factors. Experimental results demonstrate that Serom outperforms baseline systems in terms of speech quality, intelligibility, and speaker similarity, particularly under severe noisy and reverberant conditions. Evaluation on the DNS Challenge test set further shows that the proposed model generalizes effectively to unseen acoustic environments.

5. REFERENCES

- [1] Philipos C. Loizou et al., “Reasons why current speech-enhancement algorithms do not improve speech intelligibility and suggested solutions,” *IEEE TASLP*, vol. 19, no. 1, pp. 47–56, 2011.
- [2] Norbert Wiener, *Extrapolation, Interpolation, and Smoothing of Stationary Time Series*, MIT Press, 1949.
- [3] Y. Ephraim, “A minimum mean square error approach for speech enhancement,” in *ICASSP*, 1990, pp. 829–832.
- [4] Yanhui Tu et al., “2d-to-2d mask estimation for speech enhancement based on fully convolutional neural network,” in *ICASSP*, 2020, pp. 6974–6978.
- [5] Felix Weninger et al., “Speech enhancement with LSTM recurrent neural networks and its application to noise-robust ASR,” in *LVA/ICA*, 2015, pp. 91–99.
- [6] Yuxuan Wang et al., “On training targets for supervised speech separation,” *IEEE/ACM TASLP*, vol. 22, no. 12, pp. 1849–1858, 2014.
- [7] Haixin Zhao et al., “Towards complex-valued VAE-based distillation for representation learning in speech enhancement,” in *ITG Conf. Speech Commun.*, 2025, pp. 101–105.
- [8] Meng Ge et al., “Environment-dependent attention-driven recurrent convolutional neural network for robust speech enhancement,” in *Interspeech*, 2019, pp. 3153–3157.
- [9] Yen-Ju Lu et al., “Conditional diffusion probabilistic model for speech enhancement,” in *ICASSP*, 2022, pp. 7407–7411.
- [10] Haixin Zhao et al., “Study of lightweight transformer architectures for single-channel speech enhancement,” in *EUSIPCO*, 2025, pp. 101–105.
- [11] Hyungchan Song et al., “Exploring WavLM on speech enhancement,” in *SLT*, 2023, pp. 451–457.
- [12] Junyi Peng et al., “Target speech extraction with pre-trained self-supervised learning models,” in *ICASSP*, 2024, pp. 10421–10425.
- [13] Doyeon Kim et al., “Enhancing neural speech embeddings for generative speech models,” in *IWAENC*, 2024.
- [14] Xiaobin Rong et al., “PASE: Leveraging the phonological prior of WavLM for low-hallucination generative speech enhancement,” in *AAAI*, 2026, vol. 40, pp. 32826–32834.
- [15] Jixun Yao et al., “GenSE: Generative speech enhancement via language models using hierarchical modeling,” in *ICLR*, 2025.
- [16] Haoyin Yan et al., “UniSE: A unified framework for decoder-only autoregressive LM-based speech enhancement,” *arXiv:2510.20441*, 2025.
- [17] Sanyuan Chen et al., “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE JSTSP*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [18] Kuo-Hsuan Hung et al., “Boosting self-supervised embeddings for speech enhancement,” in *Interspeech*, 2022, pp. 3038–3042.
- [19] Alex Graves et al., “Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks,” in *ICML*, 2006, pp. 369–376.
- [20] Hubert Siuzdak, “Vocos: Closing the gap between time-domain and fourier-based neural vocoders for high-quality audio synthesis,” in *ICLR*, 2024.
- [21] Chandan K. A. Reddy et al., “The INTERSPEECH 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results,” in *Interspeech*, 2020.
- [22] Tom Ko et al., “A study on data augmentation of reverberant speech for robust speech recognition,” in *ICASSP*, 2017, pp. 5220–5224.
- [23] Won Jang et al., “UnivNet: A neural vocoder with multi-resolution spectrogram discriminators for high-fidelity waveform generation,” *arXiv:2106.07889*, 2021.
- [24] Rithesh Kumar et al., “High-fidelity audio compression with improved RVQGAN,” in *NeurIPS*, 2023, vol. 36, pp. 27980–27993.
- [25] Yifan Peng et al., “OWSM v3.1: Better and faster open whisper-style speech models based on E-Branchformer,” in *Interspeech*, 2024, pp. 352–356.
- [26] Jee-weon Jung et al., “Pushing the limits of raw waveform speaker recognition,” in *Interspeech*, 2022, pp. 2228–2232.
- [27] Laurens Van der Maaten et al., “Visualizing data using t-SNE,” *JMLR*, vol. 9, no. 11, 2008.